Southern New Hampshire University

DriverPass — Business Requirements Document

System Components and Design

Purpose

- Define the requirements for a secure, cloud-based system for DriverPass that helps students prepare for DMV driving tests through online learning materials, practice exams, and behind-the-wheel lesson scheduling.
- Provide a centralized system that enables customers, employees, instructors, management, and IT administrators to perform their respective tasks efficiently.
- Support DriverPass business operations by integrating customer management, scheduling, online training, and progress tracking into a single system.
- Improve communication between DriverPass staff and customers while simplifying appointment scheduling and monitoring student progress.

System Background

- DriverPass was created to address the high failure rate of DMV driving tests by providing additional driver education and behind-the-wheel training.
- The company wants to offer online training materials, practice exams, and behind-the-wheel driving lessons through a single integrated system.
- The system must support multiple user roles, including customers, the owner, office staff, instructors, and the IT administrator.
- Customers should be able to register, purchase training packages, schedule, modify, and cancel appointments, complete online coursework, and take practice exams.
- Employees need tools to manage appointments, maintain customer records, monitor lesson progress, generate reports, and keep training materials current with changes to DMV requirements.

Objectives and Goals

- Allow customers to create and manage secure accounts.
- Allow customers to schedule, modify, and cancel behind-the-wheel driving lessons online.
- Allow office staff to schedule, modify and cancel appointments on behalf of customers.
- Track instructor assignments, training vehicles, lesson schedules, and customer reservations.
- Record and monitor student progress for online training and practice exams.
- Display practice exam status, completion status, and testing history.
- Allow instructors to record lesson notes and student performance after each driving session.
- Enforce role-based access for customers, instructors, office staff, management and IT administrators.

- Provide secure password reset and account recovery capabilities.
- Maintain an audit log showing who created, modified, or canceled appointments and reservations.
- Receive updates when DMV rules, policies, or practice exam content change.
- Allow management to generate reports and export business information as needed.
- Support future expansion by allowing training packages to be added, modified, enabled or disabled without requiring major system redesign.

Requirements

Nonfunctional Requirements

Performance Requirements

- The system shall operate as a cloud-based, web application that is available to customers and employees from any location with an internet connection.
- The system shall provide responsive performance for users accessing online training, practice exams, scheduling, and account information.
- The system shall support multiple users accessing the system simultaneously without significant performance degradation.
- The system shall maintain high availability to allow customers to register, schedule lessons, and complete online training whenever needed.
- The system shall support timely updates to training materials when DMV requirements change while minimizing disruption to users.

Platform Constraints

- The system shall be accessible through modern web browsers rather than requiring installation of client software.
- The system shall support Windows, macOS, and Linux desktop operating systems.
- The system shall support mobile access from Android and iOS devices.
- The system shall use a back-end database to store customer accounts, scheduling information, lesson records, and training data.
- The system shall allow the IT administrator to maintain and update the application without requiring changes to customer devices.
- The cloud provider shall manage routine backups and infrastructure availability to reduce the amount of routine maintenance required by DriverPass IT staff.

Accuracy and Precision

- The system shall distinguish users through unique usernames or email addresses and assign permissions based on their role (customer, instructor, office staff, management, or IT administrator).
- The system shall validate user input to reduce data-entry errors and ensure required information is complete before it is saved.
- The system shall maintain accurate customer records, appointment schedules, lesson notes, and test results.
- The system shall notify the IT administrator when critical system errors, failed updates, or other issues affect normal system operation.
- The system shall prevent unauthorized users from accessing or modifying information outside their assigned role.

Adaptability

- The system shall allow the IT administrator to add, modify, disable, or remove user accounts without changing the application's source code.
- The system shall support updates to DMV training materials, practice exams, and business information without requiring major system redesign.
- The system shall remain compatible with current versions of supported web browsers and operating systems as technology evolves.
- The system shall allow training packages and other system features to be added or modified to support future business growth.
- The IT administrator shall have sufficient administrative privileges to maintain the system, manage users, perform updates, and resolve operational issues.

Security

- The system shall require users to authenticate with a valid username (or email address) and password before accessing protected features.
- The system shall use secure, encrypted communication between client devices and the cloud-hosted application to protect sensitive information.
- The system shall enforce role-based access so users can access only the information and functions appropriate for their assigned role.
- The system shall lock or temporarily disable an account after multiple unsuccessful login attempts to help protect against brute-force attacks.
- The system shall provide a secure password reset process for users who forget their password.
- The system shall maintain an audit log of significant user activities, including account changes and appointment modifications.

- The system shall protect customer information and business data from unauthorized access or modification.

Functional Requirements

- The system shall allow customers to create and manage secure user accounts.
- The system shall authenticate users before granting access to protected system features.
- The system shall allow customers to purchase DriverPass training packages online.
- The system shall display available training packages, pricing, and package details.
- The system shall allow customers to schedule, modify, and cancel behind-the-wheel driving lessons.
- The system shall allow office staff to manage appointments on behalf of customers.
- The system shall assign instructors and vehicles to scheduled lessons.
- The system shall prevent scheduling conflicts for instructors, vehicles, and appointments.
- The system shall provide customers with access to online training materials.
- The system shall provide practice exams that reflect current DMV requirements.
- The system shall record customer progress through online training and practice exams.
- The system shall allow instructors to record lesson notes and student performance.
- The system shall maintain customer records, lesson history, and appointment information.
- The system shall generate reports for management.
- The system shall notify users of appointment confirmations, cancellations, and schedule changes.
- The system shall allow the IT administrator to manage users and maintain system data.
- The system shall allow authorized personnel to update training materials when DMV requirements change.
- The system shall maintain an audit log of significant system activities.

User Interface

- Customers shall access the system through a web browser on desktop or mobile devices.
- The interface shall provide separate views appropriate for customers, instructors, office staff, management, and IT administrators.
- Customers shall be able to register, purchase training packages, access online training, complete practice exams, and manage appointments through the interface.
- Instructors shall be able to view schedules, access student information, and record lesson notes.
- Office staff shall be able to manage customer accounts, scheduling, and reservations.
- Management shall be able to access reports and monitor business operations.
- The IT administrator shall have access to user management, system maintenance, and administrative functions.
- The interface shall be intuitive and easy to navigate for users with varying levels of computer experience.

Assumptions

- Users have reliable internet access to use the cloud-based application.
- Users have access to a supported web browser on a desktop, laptop, tablet, or mobile device.
- DriverPass employees will receive training before using the system.
- The DMV will continue providing updated rules, policies, and testing information that DriverPass can incorporate into its training materials.
- DriverPass will maintain an IT administrator responsible for user management and system maintenance.
- Cloud hosting services will provide reliable system availability and routine infrastructure maintenance.

Limitations

- The system depends on an active internet connection and cannot be fully used offline.
- The accuracy of training materials depends on timely updates to DMV rules and regulations.
- The system is limited by DriverPass's available budget, staffing, and implementation schedule.
- The system cannot guarantee the accuracy of information entered by users.
- System performance depends on the availability and reliability of the cloud hosting provider.
- Future enhancements beyond the original project scope may require additional development time and resources.

Gantt Chart

